Estd:1980

# SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)
Accredited by NAAC with 'A' Grade
UG Programmes CE,CSE,ECE,EEE,IT & ME are Accredited by NBA
Chinna Amiram, Bhimavaram-534204. (AP)

| Regulation: R20 | I / IV - B.Tech. I - Semester | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| ARTIFICIAL INTELLIGENCE & DATA SCIENCE | | | | | | | | | |
| SCHEME OF INSTRUCTION & EXAMINATION (With effect from 2020-21 admitted Batch onwards) | | | | | | | | | |
| **Course Code** | **Course Name** | **Category** | **Cr.** | **L** | **T** | **P** | **Int. Marks** | **Ext. Marks** | **Total Marks** |
| B20 HS 1101 | English | HS | 3 | 3 | 0 | 0 | 30 | 70 | 100 |
| B20 BS 1101 | Mathematics-I | BS | 3 | 3 | 0 | 0 | 30 | 70 | 100 |
| B20 BS 1102 | Applied Physics | BS | 3 | 3 | 0 | 0 | 30 | 70 | 100 |
| B20 CS 1101 | Programming for Problem Solving Using C | ES | 3 | 3 | 0 | 0 | 30 | 70 | 100 |
| B20 IT 1101 | Fundamentals of Computers and Information Technology | ES | 3 | 3 | 0 | 0 | 30 | 70 | 100 |
| B20 CS 1103 | Programming for Problem Solving Using C Lab | ES | 1.5 | 0 | 0 | 3 | 15 | 35 | 50 |
| B20 BS 1107 | Applied Physics Lab | BS | 1.5 | 0 | 0 | 3 | 15 | 35 | 50 |
| B20 CS 1104 | Computer Engineering Workshop | ES | 1.5 | 0 | 0 | 3 | 15 | 35 | 50 |
| | | **TOTAL** | **19.5** | **15** | **0** | **9** | **195** | **455** | **650** |

| Course Code | Category | L | T | P | C | I.M | E.M | Exam |
|---|---|---|---|---|---|---|---|---|
| **B20HS1101** | **HS** | **3** | **--** | **--** | **3** | **30** | **70** | **3Hrs** |

## ENGLISH

(Common to AIDS,CE,CSE,ECE,EEE,IT&ME)

**Introduction:**

The course is designed to train students in receptive as well as productive skills by incorporating a comprehensive, coherent and integrated approach that improves the learners' ability to effectively use English language in academic/ workplace contexts. The shift is from *learning about the language* to *using the language*. On successful completion of the compulsory English language course/s in B.Tech., learners would be confident of appearing for international language qualification/proficiency tests such as GRE, GMAT, IELTS, TOEFL and BEC besides being able to handle the writing tasks and verbal ability components ofcampusplacementtests.Activitybasedteaching-learningmethodswouldbeadoptedto ensure that learners would engage in actual use of language both in the classroom and laboratorysessions.

**Course Objectives:**

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|---|---|
| 1. | To facilitate effective listening skills for better comprehension of varied accents spoken at national and global levels. |
| 2. | To focus on appropriate reading strategies for better comprehension of multiple texts and authentic materials. |
| 3. | To improve speaking skills through participation in activities such as role plays, discussions and structured talks/oral presentations. |
| 4. | To impart effective strategies for good writing and demonstrate the same in both summarizing and analyzing; writing well-organized essays, letters, e-mails, CV's and reports. |
| 5. | To provide knowledge of grammatical structures and vocabulary and encourage their appropriate use in speech and writing. |

**Course Outcomes:** At the end of the Course the students will be able to

| S.No | OutCome | KL |
|---|---|---|
| 1. | Identify the context, topic and pieces of specific information by understanding andresponding to the social or transactional dialogues spoken by native speakers of English. | K3 |
| 2. | Apply suitable strategies for skimming and scanning to get the main idea of a textand locate specific information. | K3 |
| 3. | Build confidence and adapt themselves to the social and public discourses,discussions and presentations. | K3 |
| 4. | Apply the principles of writing to paragraphs, arguments, essaysand formal/informal communication. | K3 |
| 5. | Construct sentences using proper grammatical structures and correct word forms. | K4 |

## SYLLABUS

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| **UNIT-I (8 Hrs)** | **Lesson: A Drawer full of happiness** from *Infotech English*, Maruthi Publications.<br>**Listening:** Listening to short audio texts and identifying the topic, context and specific pieces of information to answer a series of questions both in speaking and writing. |

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| | **Speaking:** Self- introduction and introducing others. Asking and answering general questions on topics such as home, family, work, studies and interests.<br><br>**Reading:** Skimming text to get the main idea. Scanning to look for specific pieces of information.<br><br>**Reading for Writing:** Paragraph Writing (Hints Development), general essays using suitable cohesive devices; linkers, sign posts and transition signals; mechanics of writing, punctuation.<br><br>**Vocabulary:**Technical vocabulary from across technical branches (20) GRE Vocabulary (20), antonyms and synonyms, word applications, verbal reasoning and sequencing of words.<br><br>**Grammar:** Content words and function words; parts of Speech, tenses, word order in sentences, sentence structures. |
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| **UNIT-II (8 Hrs)** | **Lesson-: Nehru's letter to his daughter, Indira on her birthday**from*Infotech English*, Maruthi Publications**.**<br>**Listening:** Answering a series of questions about main idea and supporting ideas after listening to audio texts both in speaking and writing.<br><br>**Speaking**: Discussion in pairs/ small groups on specific topics followed by short structured talks, functional English: greetings and leave takings.<br><br>**Reading:** Identifying sequence of ideas; recognizing verbal techniques that help to link the ideas in a paragraph together.<br><br>**Reading for Writing**: Identifying the main ideas, rephrasing and summarizing them (précis writing); avoiding redundancies and repetitions.<br>**Vocabulary:** Technical vocabulary from across technical branches (20 words). GRE Vocabulary Analogies (20 words), antonyms and synonyms, word applications.<br>**Grammar:** Articles, prepositions, conjunctions, use of synonyms and antonyms. |
| | |
| **UNIT-III (8 Hrs)** | **Lesson: Stephen Hawking-Positivity'Benchmark'**from*Infotech English*,Maruthi Publications.<br><br>**Listening:**Listening for global comprehension and summarizing what is listened to both in speaking andwriting.<br><br>**Speaking:** Discussing specific topics in pairs or small groups and reporting what is discussed. Functional English: complaining andapologizing.<br><br>**Reading:** Reading a text in detail by making basic inferences -recognizing: and interpreting specific context clues; strategies to use text clues for comprehension, criticalreading.<br><br>**Reading for Writing:** Letter writing- types, format and principles of letter writing, E-mail etiquette, writing a Resume/CV and covering letter.<br>**Vocabulary:** Technical vocabulary from across technical branches (20 words. GRE. Vocabulary 20 words), Idioms & Phrasal verbs, Homonyms, word applications, sequencing of words. |

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| | **Grammar:** Sentence Structures, Transformation of sentences (Active and passive Voice, Degrees of comparison, Simple, Compound and Complex). |
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| **UNIT-IV (8 Hrs)** | **Lesson: Liking a Tree, Unbowed: Wangari Maathai biography**from *Infotech English*, MaruthiPublications.<br>**Listening:** Making predictions while listening to conversations/ transactional dialogues without video (only audio), listening to audio-visual texts.<br>**Speaking:** Role plays for practice of conversational English in academic contexts (formal and informal) - asking for and giving information/directions. Functional English: asking for permissions, requesting,Inviting.<br>**Reading:** Studying the use of graphic elements in texts to convey information, reveal trends/patterns/relationships, communicative process or display complicated data.<br>**Reading for Writing:**Information transfer; describe, compare, contrast, identify significance/trends based on information provided in figures/charts/graphs/tables. Pamphlet writing, writing for media, writing SOP's.<br>**Vocabulary:**Technical vocabulary from across technical branches (20 words GRE Vocabulary (20 words), antonyms and synonyms, word applications, cloze encounters, foreign phrases.<br>**Grammar:**Quantifying expressions - adjectives and adverbs: comparing and contrasting, question Tags, direct and indirect speech, reporting for academic purposes. |
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| **UNIT-V (8 Hrs)** | **Lesson: Stay Hungry–Stay Foolish**from *Infotech English*, Maruthi Publications.<br>**Listening:** Identifying key terms, understanding concepts and interpreting the concepts both in speaking and writing.<br>**Speaking:** Formal oral presentations on topics from academic contexts– with/without the use of PPT slides. Functional English: Suggesting/Opiniongiving.<br>**Reading:** Reading for comprehension, RAP Strategy - intensive reading and extensive reading techniques.<br>**Reading for Writing:** Report writing, writing academic proposals- writing research articles: format and style.<br>**Vocabulary:** Technical vocabulary from across technical branches (20 words GRE 'Vocabulary (20 words, antonyms and synonyms, word applications, coherence, matching emotions.<br>**Grammar:** Editing short texts — identifying and correcting common errors in grammar and usage (articles, prepositions, tenses, subject-verb agreement, parallel structures, phrases and clauses). |
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| **Text Books:** | |
|---|---|
| 1 | *Infotech English*, Maruthi Publications. |
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| **Reference Books:** | |
| 1. | Bailey, Stephen. Academic writing: A Handbook for International Students. Routledge,2014. |
| 2. | Chase. Becky Tarver. Pathways: Listening, Speaking and Critical Thinking. Heinley ELT;2nd Edition, 2018. |
| 3. | Skilful Level 2 Reading & Writing Student's Book Pack (B1). Macmillan Educational. |
| 4. | Hewing, Martin. Cambridge Academic English (B2). CUP, 2012. |
| | |
| **E-Resources:** | |
| **Grammar/Listening/Writing** | |
| | 1-language.com |
| | http://www.5minuteenglish.com/ |
| | https://www.englishpractice.com/ |
| **Grammar/Vocabulary** | |
| | English Language Learning Online |
| | http://www.bbc.co.uk/learningenglish/ |
| | http://www.better-english.com/ |
| | http://www.nonstopenglish.com/ |
| | https://www.vocabulary.com/ |
| | BBC Vocabulary Games |
| | Free Rice Vocabulary Game |
| **Reading** | |
| | https://www.usingenglish.com/comprehension/ |
| | https://www.englishclub.com/reading/short-stories.htm |
| | https://www.english-online.at/ |
| **Listening** | |
| | https://learningenglish.voanews.com/z/3613 |
| | http://www.englishmedialab.com/listening.html |
| **Speaking** | |
| | https://www.talkenglish.com/ |
| | BBC Learning English – Pronunciation tips |
| | Merriam-Webster – Perfect pronunciation Exercises |
| **All Skills** | |
| | https://www.englishclub.com/ |
| | http://www.world-english.org/ |
| | http://learnenglish.britishcouncil.org/ |
| | Online Dictionaries |
| | Cambridge dictionary online |
| | MacMillan dictionary |
| | Oxford learner's dictionaries |

| Course Code | Category | L | T | P | C | I.M | E.M | Exam |
|---|---|---|---|---|---|---|---|---|
| **B20BS1101** | **BS** | **3** | **--** | **--** | **3** | **30** | **70** | **3 Hrs.** |

## MATHEMATICS-I
## (LINEAR ALGEBRA AND DIFFERENTIAL EQUATIONS)

(Common to AIDS, CE, CSE, ECE, EEE, IT & ME)

**Pre-requisites:** Calculus of functions of a single variable and Matrices.

**Course Objectives:**Students are expected to learn

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|---|---|
| 1. | Concepts of linear algebra and methods of solution of linear simultaneous algebraicequations. |
| 2. | Eigen values, Eigen vectors and quadratic forms. |
| 3. | First order ordinary differential equations and some simple geometrical and physical applications. |
| 4. | Orthogonal trajectories, Simple electrical circuits and Newton's law of cooling. |
| 5. | Methods of solution of linear higher order ordinary differential equations. |
| 6 | Concepts of Laplace transforms and their applications for solving ODE. |

**Course Outcomes:** At the end of the course the student will be able to

| S.No | Outcome | KL |
|---|---|---|
| 1. | Solve a given system of linear algebraic equations | K3 |
| 2. | Determine Eigen values and Eigen vectors of a system represented by a matrix. | K3 |
| 3. | Solve ordinary differential equations of first order and first degree. | K3 |
| 4. | Applythe knowledge in simple applications such as Newton's law of cooling, orthogonal trajectories and simple electrical circuits | K3 |
| 5. | Solve linear ordinary differential equations of second order and higher order. | K3 |
| 6. | Determine Laplace transform, inverse Laplace transform and solve linear ODE | K3 |

## SYLLABUS

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|---|---|
| **UNIT-I (10 Hrs)** | **Linear systems of equations:** Rank, Echelon form, Normal form, consistency of system of linear equations, Solution of linear systems by Gauss elimination, Jacobi and Gauss-Seidel methods. |
| **UNIT-II (10 Hrs)** | **Eigen values - Eigen vectors and Quadratic forms:** Eigen values, Eigen vectors, Properties, Cayley-Hamilton theorem, Inverse and powers of a matrix using Cayley-Hamilton theorem, Reduction to diagonal form, Quadratic forms, Reduction of a Quadratic form to Canonical form. |
| **UNIT-III (10 Hrs)** | **Differential equations of first order and first degree:** Linear, Bernoulli, Exact, Reducible to exact types. Applications: Orthogonal trajectories, Newton's Law of cooling, Simple electrical circuits.(R-L and R-C circuits only) |
| **UNIT-IV (8 Hrs)** | **Linear differential equations of higher order:** Linear Non-homogeneous equations of higher order with constant coefficients with source (RHS) term of the type $e^{ax}$, sin ax, cos ax, polynomials in x, $e^{ax}$ V(x), x V(x). Simultaneous differential equations with constant coefficients, Method of Variation of parameters. |

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|---|---|
| **UNIT-V (12 Hrs)** | **Laplace transformation:**<br>Laplace transforms of standard functions, properties,transforms of tf(t), f(t)/t, transforms of derivatives and integrals, transforms of unit step function, Dirac delta function; Inverse Laplace transforms, convolution theorem (without proof).<br>Applications: Solving ordinary differential equations (initial value problems) using Laplace transforms. |

**Text Books:**

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|---|---|
| 1. | B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers. |
| 2. | B. V. Ramana,Higher Engineering Mathematics, 2007 Edition, Tata Mc. Graw Hill Education. |
| 3. | N.P.Bali&Manish Goyal, Engineering Mathematics, Lakshmi Publications. |

**Reference Books:**

| | |
|---|---|
| 1. | V. Ravindranath&P. Vijayalakshmi, Mathematical Methods, Himalaya Publishing House. |
| 2. | Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India. |
| 3. | Michael Greenberg, Advanced Engineering Mathematics, 9th edition, Pearson. |
| 4. | Dean G. Duffy, Advanced engineering mathematics with MATLAB, CRC Press. |
| 5. | Peter O'Neil, Advanced Engineering Mathematics, Cengage Learning. |
| 6. | Srimanta Pal, Subodh C.Bhunia, Engineering Mathematics, Oxford University Press. |
| 7. | Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi. |

| Course Code | Category | L | T | P | C | I.M | E.M | Exam |
|---|---|---|---|---|---|---|---|---|
| **B20BS1102** | **BS** | **3** | **--** | **--** | **3** | **30** | **70** | **3 Hrs.** |

## APPLIED PHYSICS

(Common to AIDS, CE, EEE & ME)

**Course Objectives:**

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|---|---|
| 1. | **Impart** the knowledge in basic concepts of wave optics through thePhenomena of interference and diffraction, basic concepts and properties of dielectric and magnetic materials and semiconductors. |
| 2. | **Familiarize** the student with modern technologies like lasers, optical fibers and ultrasonics with an understanding of the science behind. |
| 3. | **Impart** the elementary concepts of nanomaterials and their significance in different engineering branches. |

**Course Outcomes:**At the end of the course the student will be able to

| S.No | Outcome | KL |
|---|---|---|
| 1. | **Interpret** the behavior of light radiation in interference and diffraction Phenomena and their applications. | K3 |
| 2. | **Explain** the classification and properties of dielectric and magnetic materials suitable for engineering applications. | K3 |
| 3. | **Understand** the basics of modern optical technologies like lasers and optical fibers and their utility in various fields. | K3 |
| 4. | **Explain** the important aspects of semiconductors and electrical conductivity in them. | K3 |
| 5. | **Understand** the basics of technology of Ultrasonics in various fields and demonstrate the synthesis and applications of nanomaterials. | K3 |

## SYLLABUS

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|---|---|
| **UNIT-I (10 Hrs)** | **WAVE OPTICS**<br>**Interference:** Principle of super position. Interference of light, interference in thin films (reflected light) – Wedge film and Newton`s rings – Applications<br>**Diffraction:** Types of diffraction, Fraunhoffer diffraction at a single alit, Diffraction grating, grating spectrum. Missing order, Resolving power, Rayleigh's Criterion, Resolving power of Grating |
| **UNIT-II (10 Hrs)** | **DIELECTRICS AND MAGNETICS**<br>**Dielectrics** : Introduction to dielectrics, Electric Polarization, Dielectric polarizability, Susceptibility, Dielectric constant, Types of Polarization, Frequency dependence of Polarization, Internal field in a dielectric, Claussius and Mosotti equation, Applications of dielectrics.<br>**Magnetics**: Introduction to magnetics, Magnetic dipole moment , Magnetization, Magnetic susceptibility and Permeability, Origin of permanent magnetic moment, Classification of magnetic materials ( Dia , Para, Ferro, Antiferro and ferri), Hysteresis – Weiss Domain theory – Ferrites, soft and hard magnetic materials, Magnetic device applications. |

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| **UNIT-III (10 Hrs)** | **LASERS AND FIBER OPTICS**<br>**Lasers**: Introduction, Interaction of radiation with matter, condition for light amplification, Einstein's relations. Requirements of lasers device Types of lasers, Design and working of Ruby and He – Ne lasers, Laser characteristics and applications.<br>**Fiber Optics**: Introduction to optical fibers, Principle of light propagation in fiber, Acceptance angle, Numerical aperture, Modes of propagations, types of fibers, classification of fibers based on refractive index profile, applications of fibers with emphasis on fiber optic communication. |
| | |
| **UNIT-IV (9 Hrs)** | **SEMICONDUCTORS**<br>Introduction, intrinsic semi conductors, density of charge carries, Fermi energy, Electrical conductivity – Extrinsic semi conductors – P-type and N-type, Density of charge carriers, dependence of Fermi energy on carrier concentration and temperature, direct and indirect band – gap semi conductors, Hall effect, Applications of Hall effect. Drift and diffusion currents, Continuity equation, applications of semi conductors. |
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| **UNIT-V (9 Hrs)** | **ULTRASONICS AND NANOMATERIALS**<br>**Ultrasonics:** Introduction, Production of Ultrasonics – Piezoelectric and Magnetostriction methods, detection of ultrasonics, acoustic grating - determination of wavelength and velocity of ultrasonics, applications of ultrasonics.<br>**Nanomaterials:** Introduction, salient features of Nanomaterials, Synthesis methods – Ball milling, Condensation, Chemical Vapour Deposition and Sol – Gel methods, Characterization techniques for nano materials - The scanning tunneling microscopy (STM) and The atomic force microscopy (AFM), Carbon nanotubes (CNTS), Applications of Nano materials. |
| | |
| **Text Books:** | |
| 1. | A text Book of Engineering Physics – M.N. Avadhanulu and P.G.Kshirasagar.-S.Chand Publications 2017 |
| 2. | Engineering Physics by HK Malik and A.K.Singh. McGrawhill Publishing Company Ltd. |
| 3. | Engineering Physics by V.Rajendran. McGrawhill Education (India)Pvt Ltd. |
| | |
| **Reference Books:** | |
| 1. | Introduction to Solid State Physics by Charles Kittel , Wiley Publications 2011 |
| 2. | Semiconductors Devices – Physics and Technology by S.M.Sze , Wiley Publications 2008 |
| 3. | Text book of Nano Science and Nano technology by TataMcGrawhill 2013. |
| 4. | Optical fiber communications by Gerd Keiser, Tata McGraw hill 2008. |
| | |
| **e-Resources:** | |
| 1. | http://library.iiti.ac.in/ |
| 2. | https://onlinecourses.nptel.ac.in/ |

| Course Code | Category | L | T | P | C | I.M | E.M | Exam |
|---|---|---|---|---|---|---|---|---|
| **B20CS1101** | **ES** | **3** | **--** | **--** | **3** | **30** | **70** | **3 Hrs.** |

## PROGRAMMING FOR PROBLEM SOLVING USING C

(Common to AIDS, CSE, ECE & IT)

**Course Objectives:**

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|---|---|
| 1. | To learn about the computer systems, computing environments, developing of a computer program, Structure of a C Program and to evaluate expressions |
| 2. | To gain knowledge of the operators, selection, control statements and repetition in C |
| 3. | To learn about the design concepts of arrays, strings, enumerated structure and union types and their usage. |
| 4. | To understand the concepts of pointers, dynamic memory allocation and know the significance of Preprocessor. |
| 5. | To learn about various File I/O operations and significance of functions |

**Course Outcomes:**At the end of the course the students will be able to

| S.No | Outcome | KL |
|---|---|---|
| 1. | Apply Precedence and Associativity rules to evaluate Expressions. | K3 |
| 2. | Make use of Decision Making and Looping statements to solve various problems in C | K3 |
| 3. | Illustrate the importance of Arrays and Strings and to apply various operations on them. | K2 |
| 4. | Solve various problems by making use of Structure and Union concepts | K3 |
| 5. | Design and implement programs to analyze the different pointer applications | K3 |
| 6. | Develop programs using Functions and Pointers. | K3 |

## SYLLABUS

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|---|---|
| **UNIT-I (10 Hrs)** | Introduction to Computers: Creating and running Programs, Computer Numbering System, Storing Integers, Storing Real Numbers Introduction to the C Language: Background, C Programs, Identifiers, Types, Variable, Constants, Input/output, Programming Examples, Scope, Storage Classes and Type Qualifiers. Structure of a C Program: Expressions Precedence and Associativity, Side Effects, Evaluating Expressions, Type Conversion Statements, Simple Programs, Command Line Arguments. |
| **UNIT-II (10 Hrs)** | Bitwise Operators: Exact Size Integer Types, Logical Bitwise Operators, Shift Operators. Selection & Making Decisions: Logical Data and Operators, Two Way Selection, Multiway Selection, More Standard Functions. Repetition: Concept of Loop, Pretest and Post-test Loops, Initialization and Updating, Event and Counter Controlled Loops, Loops in C, Other Statements Related to Looping, Looping Applications, Programming Examples. |
| **UNIT-III (10 Hrs)** | Arrays: Concepts, Using Array in C, Array Application, Two Dimensional Arrays, Multidimensional Arrays, Programming Example – Calculate Averages Strings: String Concepts, C String, String Input / Output Functions, Arrays of Strings, String Manipulation Functions String/ Data Conversion, A Programming Example – Morse Code Enumerated, Structure, and Union: The Type Definition (Type def), Enumerated Types, Structure, Unions, and Programming Application. |

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| **UNIT-IV (10 Hrs)** | Pointers: Introduction, Pointers to pointers, Compatibility, L value and R value Pointer Applications: Arrays, and Pointers, Pointer Arithmetic and Arrays, Memory Allocation Function, Array of Pointers, Programming Application. Processor Commands: Processor Commands. |
| | |
| **UNIT-V (10 Hrs)** | Functions: Designing, Structured Programs, Function in C, User Defined Functions, Inter Function Communication, Standard Functions, Passing Array to Functions, Passing Pointers to Functions, Recursion Text Input / Output: Files, Streams, Standard Library Input / Output Functions, Formatting Input / Output Functions, Character Input / Output Functions Binary Input / Output: Text versus Binary Streams, Standard Library, Functions for Files, Converting File Type. |

**Text Books:**

| | |
|---|---|
| 1. | Programming for Problem Solving, Behrouz A. Forouzan, Richard F.Gilberg, CENGAGE |
| 2. | The C Programming Language, Brian W.Kernighan, Dennis M. Ritchie, 2e, Pearson |

**Reference Books:**

| | |
|---|---|
| 1. | Computer Fundamentals and Programming, Sumithabha Das, Mc Graw Hill. |
| 2. | Programming in C, Ashok N. Kamthane, Amit Kamthane, Pearson. |
| 3. | Computer Fundamentals and Programming in C, Pradip Dey, Manas Ghosh, OXFORD. |

**e-Resources:**

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|---|---|
| 1. | https://www.geeksforgeeks.org/c-programming-language/ |
| 2. | https://www.learn-c.org/ |
| 3. | https://www.w3resource.com/c-programming-exercises/ |

| Course Code | Category | L | T | P | C | I.M | E.M | Exam |
|---|---|---|---|---|---|---|---|---|
| **B20IT1101** | **ES** | **3** | **--** | **--** | **3** | **30** | **70** | **3 Hrs.** |

## FUNDAMENTALS OF COMPUTERS AND INFORMATION TECHNOLOGY

(Common to AIDS & IT)

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|---|---|
| 1. | To understand of basic concepts of computer science and Information Technology. |
| 2. | To understand the fundamentals of hardware, software and programming. |
| 3. | To understand introductory skills relating to IT basics, computer applications, programming, interactive medias, Computer Networks, Security, Internet basics etc. |
| 4. | To Identify how software and hardware work together to perform computing tasks and how software is developed and upgraded |
| 5. | To Understand basic concepts of Data base Management systems and future scope of IT. |

**Course Outcomes:** At the end of the course the student will be able to

| S.No | Outcome | KL |
|---|---|---|
| 1. | Understand basic concepts and terminology of information technology. | K2 |
| 2. | Have a basic understanding of personal computers, related devices and their operations. | K2 |
| 3. | Understand the difference between an operating system and an application program, and what each is used for in a computer. | K2 |
| 4. | Understand Basic terminology and concepts related to Computer networks, Security. | K2 |
| 5. | Understand basics of Database management systems and learn to create forms using MS access Wizard, and current and future trends of IT. | K2 |

## SYLLABUS

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|---|---|
| **UNIT-I (6Hrs)** | **Computer Basics & Organization:** Introduction, Evolution of computers, Generation of computers, Computer Concepts, Computer System, Applications of Computers. Introduction, Central Processing Unit, Internal Communications, Machine cycle, The Bus, Instruction Set. |
| **UNIT-II (10 Hrs)** | **Computer Architecture & I/O Devices**: Memory & Storage systems: Memory Representation, RAM, ROM, Magnetic Storage Systems, Optical Storage Systems, Solid-State Storage Devices**. Input and Output Devices**: Keyboard, Pointing Devices, Scanning Devices, Optical Recognition Devices, Data Acquisition Sensors, Media Input Devices, Display Monitors, Printers, Plotters, Terminals. |

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| **UNIT-III (12Hrs)** | **Computer Software and Programming Languages**: Introduction, Types of Computer Software, System Management Programs, System Development Programs, Standard Application Programs, Unique Application Programs, Problem Solving, Structuring the Logic, Using the computer. **Operating Systems (OS)**: Introduction, History of OS, Functions of OS, Types of Operating Systems, Popular Operating Systems. **Microsoft Software**: MS-DOS, MS Word System, MS Excel System, MS Power point System, MS Access System. **Programming Languages**: History of Programming Languages, Generations of Programming Languages, Characteristics of Good Programming Language, Categorizations of High Level Languages, Popular High level Languages, Developing a Program, Running a Program. |
| | |
| **UNIT-IV (10 Hrs)** | **Data Communications, Networks, Internet and Security:** Introduction, Data communication using MODEMS, Computer Network, Network Protocols and Software, Network Devices, **Internet**: Introduction, Evolution of Internet, Basic Internet Terms, Internet applications: E-mail, FTP, Telnet, Internet Relay Chat (IRC), Chatting and instant messaging, Internet Telephony, Video Conferencing, Commerce Through Internet. **Security**: Introduction, Computer Security definition, Malicious Programs: Virus, Other destructive Programs, Affecting and Protecting Computer Systems, Cryptography, Digital Signature, Firewall. |
| | |
| **UNIT-V (10Hrs)** | **Database Fundamentals**: Introduction, Database Definition, Logical Data Concepts, Physical Data Concepts, Database Management System, DBMS Architecture, database Models, **MS Access**: Introduction, Starting with Access database, Working with Tables, Creating Forms using Wizard. **Current and Future Trends in IT**: Introduction, Electronic Commerce, Electronic Data Interchange, Smart Card, Radio frequency Identification (RFID), Brain computer Interface(BCI), Imminent technologies. |

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|---|---|
| **Text Books:** | |
| 1. | Fundamentals of Computers , E Balagurusamy, MGH ( Unit, 1,2,3 and 4(Up to Network Devices) |
| 2. | Introduction to Information technology, ITL Education Solutions Limited, 2nd edition, Pearson ( Unit 4 ( Internet and Security),Unit 5) |
| | |
| **Reference Books:** | |
| 1. | Deborah Morley and Charles S. Parker; Fundamentals of Computers; Cengage Learning, India edition; 2009. |
| 2. | Alexis Leon and Mathews Leon; Fundamentals of Information Technology; Vikas Publication, |
| 3. | Computer Fundamentals by P.K.Sinha&Preetisinha |
| 4. | Snyder, Lawrence. 2007. Fluency with information technology: Skills, concepts, and capabilities. 3nd Edition. Boston: Addison-Wesley. ISBN: 0321512391. |

| Course Code | Category | L | T | P | C | I.M | E.M | Exam |
|---|---|---|---|---|---|---|---|---|
| B20CS1103 | ES | 0 | 0 | 3 | 1.5 | 15 | 35 | 3 Hrs. |

## PROGRAMMING FOR PROBLEM SOLVING USING C LAB

Common to AIDS, CSE, ECE & IT

**Course Objectives:**

| | |
|---|---|
| 1. | Apply the principles of C language in problem solving. |
| 2. | To design & develop of C programs using Arrays, Strings, Structures, Unions and Pointers |
| 3. | To perform the file operations, preprocessor commands |
| 4. | To solve various complex problem by applying modular programming skills |

**Course Outcomes:** At the end of the course students will be able to

| S.No | Out Come | KL |
|---|---|---|
| 1. | Write, Trace and Debug the programs and correct syntax and logical errors. | K4 |
| 2. | Solve various Problems by making use of Arrays, Strings, Structures, Unions and Pointers | K3 |
| 3. | Solve a complex problem by decomposing into several modules by using Functions | K4 |
| 4. | Apply various File I/O operations | K3 |

## LIST OF PROGRAMS

| | |
|---|---|
| 1 | Exercise 1:<br>1. Write a C program to print a block F using hash (#), where the F has a height of six characters and width of five and four characters.<br>2. Write a C program to compute the perimeter and area of a rectangle with a height of 7 inches and width of 5 inches.<br>3. Write a C program to display multiple variables. |
| 2 | Exercise 2:<br>1. Write a C program to calculate the distance between the two points.<br>2. Write a C program that accepts 4 integers p, q, r, s from the user where r and s are positive and p is even. If q is greater than r and s is greater than p and if the sum of r and s is greater than the sum of p and q print "Correct values", otherwise print "Wrong values". |
| 3 | Exercise 3:<br>1. Write a C program to convert a string to a long integer.<br>2. Write a program in C which is a Menu-Driven Program to compute the area of the various geometrical shape.<br>3. Write a C program to calculate the factorial of a given number. |
| 4 | Exercise 4:<br>1. Write a program in C to display the n terms of even natural number and their sum.<br>2. Write a program in C to display the n terms of harmonic series and their sum. 1 + 1/2 + 1/3 + 1/4 + 1/5 ... 1/n terms.<br>3. Write a C program to check whether a given number is an Armstrong number or not. |
| 5 | Exercise 5:<br>1. Write a program in C to print all unique elements in an array.<br>2. Write a program in C to separate odd and even integers in separate arrays.<br>3. Write a program in C to sort elements of array in ascending order. |

| | |
|---|---|
| 6 | Exercise 6:<br>1. Write a program in C for multiplication of two square Matrices.<br>2. Write a program in C to find transpose of a given matrix. |
| 7 | Exercise 7:<br>1. Write a program in C to search an element in a row wise and column wise sorted matrix.<br>2. Write a program in C to print individual characters of string in reverse order. |
| 8 | Exercise 8:<br>1. Write a program in C to compare two strings without using string library functions.<br>2. Write a program in C to copy one string to another string. |
| 9 | Exercise 9:<br>1. Write a C Program to Store Information Using Structures with Dynamically Memory Allocation<br>2. Write a program in C to demonstrate how to handle the pointers in the program. |
| 10 | Exercise 10:<br>1. Write a program in C to demonstrate the use of & (address of) and *(value at address) operator.<br>2. Write a program in C to add two numbers using pointers |
| 11 | Exercise 11:<br>1. Write a program in C to add numbers using call by reference.<br>2. Write a program in C to find the largest element using Dynamic Memory Allocation |
| 12 | Exercise 12:<br>1. Write a program in C to swap elements using call by reference.<br>2. Write a program in C to count the number of vowels and consonants in a string using a pointer. |
| 13 | Exercise 13:<br>1. Write a program in C to show how a function returning pointer.<br>2. Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using malloc( ) function |
| 14 | Exercise 14:<br>1. Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using calloc( ) function. Understand the difference between the above two programs<br>2. Write a program in C to convert decimal number to binary number using the function. |
| 15 | Exercise 15:<br>1. Write a program in C to check whether a number is a prime number or not using the function.<br>2. Write a program in C to get the largest element of an array using the function. |
| 16. | Exercise 16:<br>1. Write a program in C to append multiple lines at the end of a text file.<br>2. Write a program in C to copy a file in another name.<br>3. Write a program in C to remove a file from the disk. |

**Reference Books:**

| | |
|---|---|
| 1. | Programming for Problem Solving, Behrouz A. Forouzan, Richard F.Gilberg, CENGAGE |
| 2. | The C Programming Language, Brian W.Kernighan, Dennis M. Ritchie, 2e, Pearson |

**e-Resources:**

| | |
|---|---|
| 1. | https://www.geeksforgeeks.org/c-programming-language/ |
| 2. | https://www.learn-c.org/ |
| 3. | https://www.tutorialspoint.com/cprogramming/index.html |

| Course Code | Category | L | T | P | C | I.M | E.M | Exam |
|---|---|---|---|---|---|---|---|---|
| **B20BS1107** | **BS** | **--** | **--** | **3** | **1.5** | **15** | **35** | **3 Hrs.** |

## APPLIED PHYSICS LAB

(Common to AIDS, CE,EEE & ME)

**Course Objectives:**

| | |
|---|---|
| 1. | To impart hands-on experience to the students entering engineering / Technology education about handling sophisticated equipment / instruments. |
| 2. | To make the students understand the theoretical aspects of various phenomena experimentally. |

**Course Outcomes:**At the end of the course students willbe able to

| S.No | **Out Come** | **KL** |
|---|---|---|
| 1. | Get hands on experience in setting up experiments and using the instruments / equipment individually. | K3 |
| 2. | Get introduced to using new / advanced technologies and understand their significance. | K3 |

## LIST OF EXPERIMENTS

| | |
|---|---|
| 1 | Determination of the Wavelength of light from a source – Diffraction Grating – Normal incidence. |
| 2 | Determination of radius of curvature of Plano convex lens – Newton's Rings. |
| 3 | Determination of the thickness of a thin spacer using interference – Air Wedge method. |
| 4 | Determination of Magnetic field along the axis of a current carrying coil –Stewart and Gee's apparatus. |
| 5 | Verification of Laws of series and parallel combinations of resistances – Carey Foster's bridge. |
| 6 | Determination of Temperature Coefficient of Resistance of a thermistor |
| 7 | To study the characteristics of PN Junction diode |
| 8 | To determine the Numerical aperture of a given optical fiber and hence to find its acceptance angle. |
| 9 | Determination of Planck constant |
| 10 | Determination of the Rigidity modulus of elasticity of a material – Torsional pendulum. |
| 11 | Verification of the laws of vibrations in stretched stings - Sonometer. |
| 12 | Determination of the frequency of the AC supply – AC Sonometer. |
| 13 | To determine refractive indices ($\mu$o and $\mu$e) of a birefringent material (prism). |

**Reference Books:**


| | |
|---|---|
| 1. | Advanced Practical Physics Vol 1& 2 SP Singh & M.S Chauhan Pragati Prakashan ,Meerut |

| Course Code | Category | L | T | P | C | I.M | E.M | Exam |
|---|---|---|---|---|---|---|---|---|
| **B20CS1104** | **ES** | **--** | **--** | **3** | **1.5** | **15** | **35** | **3 Hrs.** |

## COMPUTER ENGINEERING WORKSHOP

(Common to AIDS, CSE & IT)

**Course Objectives:** Skills and knowledge provided by this subject are the following:

| | |
|---|---|
| 1. | PC Hardware: Identification of basic peripherals, Assembling a PC, Installation of system software like MS Windows, device drivers, etc. Troubleshooting of PC Hardware and Software issues. |
| 2. | Internet & World Wide Web: Different ways of hooking the PC on to the internet from home and workplace and effectively usage of the internet, web browsers, email, newsgroups and discussion forums. Awareness of cyber hygiene (protecting the personal computer from getting infected with the viruses), worms and other cyber-attacks. |
| 3. | Productivity Tools: Understanding and practical approach of professional word documents, excel spread sheets, power point presentations and personal web sites using the Microsoft suite office tools. |

**Course Outcomes:** At the end of the course students will be able to

| S.No | Out Come | KL |
|---|---|---|
| 1 | Identify, assemble and update the components of a computer | K3 |
| 2 | Configure, evaluate and select hardware platforms for the implementation and execution of computer applications, services and systems | K5 |
| 3 | Make use of tools for converting pdf to word and vice versa | K3 |
| 4 | Develop presentation, documents and small applications using productivity tools such as word processor, presentation tools, spreadsheets, HTML, LaTex | K3 |

## SYLLABUS

**Note: Faculty to consolidate the workshop manuals using the textbook and references**

**List of Exercises:**

| | |
|---|---|
| **Task 1** | **Identification of the peripherals of a computer -** Prepare a report containing the block diagram of the computer along with the configuration of each component and its functionality. Describe about various I/O Devices and its usage. |
| **Task 2:** | Practicing disassembling and assembling components of a PC |
| **Task 3:** | Installation of Device Drivers, MS Windows, Linux Operating systems and Disk Partitioning, dual boating with Windows and Linux |
| **Task 4:** | Introduction to Memory and Storage Devices, I/O Port, Assemblers, Compilers, Interpreters, Linkers and Loaders. |
| **Task 5:** | Demonstration of Hardware and Software Troubleshooting |
| **Task 6**: | Demonstrating Importance of Networking, Transmission Media, Networking Devices-Gateway, Routers, Hub, Bridge, NIC, Bluetooth Technology, Wireless Technology, Modem, DSL, and Dialup Connection. |

| | |
|---|---|
| **Task 7** | Surfing the Web using Web Browsers, Awareness of various threats on the Internet and its solutions, Search engines and usage of various search engines, Need of anti-virus, Installation of anti-virus, configuring personal firewall and windows update.<br>(Students should get connected to their Local Area Network and access the Internet. In the process they should configure the TCP/IP setting and demonstrate how to access the websites and email. Students customize their web browsers using bookmarks, search toolbars and pop up blockers) |
| **Task 8:** | **Productivity Tools:**<br>Basic HTML tags, Introduction to HTML5 and its tags, Introduction to CSS3 and its properties. Preparation of a simple website/ homepage,<br>Assignment: Develop your home page using HTML Consisting of your photo, name, address and education details as a table and your skill set as a list.<br>Features to be covered:- Layouts, Inserting text objects, Editing text objects, Inserting Tables, Working with menu objects, Inserting pages, Hyper linking, Renaming, deleting, modifying pages,etc., |
| **Task 9**: | Demonstration and Practice of various features of Microsoft Word Assignment:<br>1. Create a project certificate.<br>2.Creating a newsletter<br>Features to be covered:-Formatting Fonts, Paragraphs, Text effects, Spacing, Borders and Colors, Header and Footer, Date and Time option, tables, Images, Bullets and Numbering, Table of Content, Newspaper columns, Drawing toolbar and Word Art and Mail Merge in word etc., |
| **Task 10:** | Demonstration and Practice of various features MicrosoftExcel<br>Assignment: 1. Creating a scheduler<br>2.CalculatingGPA<br>3.Calculating Total, average of marks invarious subjects and ranks of Students based on marks<br>Features to be covered:- Format Cells, Summation, auto fill,Formatting Text, Cell Referencing, Formulae in excel, Charts, Renaming and Inserting worksheets,etc., |
| **Task 11:** | Demonstration and Practice of various features Microsoft Power Point<br>Features to be covered:- Slide Layouts, Inserting Text, Word Art, Formatting Text, Bullets and Numbering, Auto Shapes, Hyperlinks Tables and Charts, Master Layouts, Types of views, Inserting – Background, textures, Design Templates, etc., |
| **Task 12:** | Demonstration and Practice of various features LaTeX – document preparation, presentation (Features covered in Task 9 and Task 11 need to be explored in LaTex) |
| **Task 13:** | Tools for converting word to pdf and pdf to word |
| **Task14:** | Internet of Things (IoT): IoT fundamentals, applications, protocols, communication models, architecture, IoT devices. |

**Text Books:**

| | |
|---|---|
| 1 | Computer Fundamentals, Anita Goel, Pearson India Education,2017 |
| 2 | PC Hardware Trouble Shooting Made Easy,TMH |
| 3 | Introduction to Information Technology, ITL Education Solutions Limited,2ndEdition, Pearson, 2020 |
| 4 | Upgrading and Repairing PCs, 18th Edition, Scott Mueller, QUE, Pearson,2008 |

| | |
|---|---|
| 5 | LaTeX Companion – Leslie Lamport,PHI/Pearson |
| 6 | Introducing HTML5, Bruce Lawson, Remy Sharp, 2nd Edition, Pearson,2012 |
| 7 | Teach yourself HTML in 24 hours, ByTechmedia |
| 8 | HTML 5 and CSS 3.0 to the Real World by Alexis Goldstein, Sitepointpublication. |
| 9 | Internet of Things, Technologies, Applications, Challenges and Solutions, B K Tripathy, J Anuradha, CRCPress |
| 10 | Comdex Information Technology Course Tool Kit, Vikas Gupta, WileyDreamtech. |
| 11 | IT Essentials PC Hardware and Software Companion Guide Third Edition by David Anfinson and Ken Quamme, CISCO Press, PearsonEducation. |
| 12 | Essential Computer and IT Fundamentals for Engineering and Science Students,Dr.N.B. Venkateswarlu, S. Chand Publishers |

| Regulation: R20 | I / IV - B.Tech. II - Semester | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| ARTIFICIAL INTELLIGENCE & DATA SCIENCE | | | | | | | | | |
| SCHEME OF INSTRUCTION & EXAMINATION<br>(With effect from 2020-21 admitted Batch onwards) | | | | | | | | | |
| **Course Code** | **Course Name** | **Category** | **Cr** | **L** | **T** | **P** | **Int. Marks** | **Ext. Marks** | **Total Marks** |
| B20 BS 1201 | Mathematics-II | BS | 3 | 3 | 0 | 0 | 30 | 70 | 100 |
| B20 BS 1203 | Applied Chemistry | BS | 3 | 3 | 0 | 0 | 30 | 70 | 100 |
| B20 IT 1201 | Digital Logic Design | ES | 3 | 3 | 0 | 0 | 30 | 70 | 100 |
| B20 IT 1202 | Object Oriented Programming through C++ | ES | 3 | 3 | 0 | 0 | 30 | 70 | 100 |
| B20 EE 1203 | Principles of Electrical and Electronics Engineering | ES | 3 | 3 | 0 | 0 | 30 | 70 | 100 |
| B20 BS 1208 | Applied Chemistry Lab | BS | 1.5 | 0 | 0 | 3 | 15 | 35 | 50 |
| B20 HS 1202 | Communication Skills Lab | HS | 1.5 | 0 | 0 | 3 | 15 | 35 | 50 |
| B20 IT 1203 | Object Oriented Programming through C++ Lab | ES | 1.5 | 0 | 0 | 3 | 15 | 35 | 50 |
| B20 MC 1201 | Environmental Science | MC | 0 | 2 | 0 | 0 | -- | -- | -- |
| B20 MC 1203 | National Service Scheme (NSS) | MC | 0 | 0 | 0 | 2 | -- | -- | -- |
| **TOTAL** | | | **19.5** | **17** | **0** | **11** | **195** | **455** | **650** |

| Course Code | Category | L | T | P | C | I.M | E.M | Exam |
|---|---|---|---|---|---|---|---|---|
| **B20BS1201** | **BS** | **3** | **--** | **--** | **3** | **30** | **70** | **3 Hrs.** |

**MATHEMATICS – II**

**(FOURIER ANALYSIS AND PARTIAL DIFFERENTIAL EQUATIONS)**

(Common to AIDS, CE, CSE, ECE, EEE, IT & ME)

**Prerequisites:**Calculus of functions of a single variable and Geometry

**Course Objectives:** Students are expected to learn:

| | |
|---|---|
| 1. | How to expand an aperiodic function in a Fourier series. |
| 2. | How to find Fourier transform for a given function and evaluate some real definite integrals. |
| 3. | Application of partial differentiation for determining maxima/ minima of functions. |
| 4. | Evaluation of real definite integrals. |
| 5. | Formation and solution of linear partial differential equations |
| 6. | Solution of one-dimensional wave equation and one-dimensional heat equation by the method of separation of variables. |

**Course Outcomes:** At the end of the course students will be able to

| S. No | Outcome | KL |
|---|---|---|
| 1. | DetermineFourier series and half range series of functions | K3 |
| 2. | Determine Fouriertransforms of non-periodic functions and also use them to evaluate integrals. | K3 |
| 3. | Compute partial derivatives, total derivative and Jacobians. | K3 |
| 4. | Find maxima/minima of functions of two variables and evaluate some real definite integrals. | K3 |
| 5. | Form partial differential equations and solve Lagrange linear equation. Solve linear higher order homogeneous and non-homogeneous PDEs. | K3 |
| 6. | Find theoretical solution of one-dimensional wave equation and one-dimensional heat equation | K3 |

**SYLLABUS**

| | |
|---|---|
| **UNIT-I (10 Hrs)** | **Fourier Series**<br>Introduction, Periodic functions, Fourier series of a periodic function, Dirichlet's conditions, Change of interval.<br>Even and odd functions, Half-range sine and cosine series. |
| **UNIT-II (12 Hrs)** | **Fourier Transforms**<br>Fourier integral theorem (without proof), Complex form of Fourier integral, Fourier sine and cosine integrals, Fourier transform, Fourier sine and cosine transforms, Finite Fourier transforms, properties, inverse transforms, Parseval's Identities. |
| **UNIT-III (10 Hrs)** | **Partial differentiation:**<br>Introduction, Homogeneous functions, Euler's theorem, Chain rule, Total derivative, Jacobians and their properties.<br>Applications: Taylor series expansion for a function of two variables, Maxima and Minima of functions of two variables with and without constraints, Lagrange's method.Leibnitz's rules for differentiation under integral sign. |

| | |
|---|---|
| **UNIT-IV (10 Hrs)** | **First order and higher order partial differential equations:** Formation of partial differential equations by elimination of arbitrary constants and arbitrary functions, solutions of Lagrange linear equation.Solutions of Linear homogeneous and non-homogeneous partial differential equations with constant coefficients –source (RHS) terms of the type $e^{ax+by}$, sin (ax+by), cos (ax+by), $x^m y^n$. |
| | |
| **UNIT-V (10 Hrs)** | **Applications of partial differential equations:** Method of separation of variables, One –dimensional wave equation, the D'Alembert's solution, one- dimensional heat equation |
| | |

**Text Books:**

| | |
|---|---|
| 1. | B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers. |
| 2. | N.P.Bali& Manish Goyal, A Text book of Engineering Mathematics, Lakshmi Publications. |
| 3. | B. V. Ramana, Higher Engineering Mathematics, 2007 Edition, Tata Mc. Graw Hill Education. |

**Reference Books:**

| | |
|---|---|
| 1. | Dean G. Duffy, Advanced engineering mathematics with MATLAB, CRC Press. |
| 2. | V.Ravindranath and P. Vijayalakshmi, Mathematical Methods, Himalaya Publishing House. |
| 3. | Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India. |
| 4. | David Kincaid, Ward Cheney, Numerical Analysis-Mathematics of Scientific Computing, 3rd Edition, Universities Press. |
| 5. | Srimanta Pal, Subodh C.Bhunia, Engineering Mathematics, Oxford University Press. |
| 6. | Dass H.K., RajnishVerma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi. |

| Course Code | Category | L | T | P | C | I.M | E.M | Exam |
|---|---|---|---|---|---|---|---|---|
| B20BS1203 | BS | 3 | -- | -- | 3 | 30 | 70 | 3Hrs |

## APPLIED CHEMISTRY

(Common to AIDS,CE,EEE & ME)

**Course Objectives:**

| | |
|---|---|
| 1. | To understand the physical and mechanical properties of Polymers/Plastics/elastomers helps in selecting suitable materials for different purpose. |
| 2. | To create awareness on fuels as a source of energy for industries like thermal power stations, steel industry, fertilizer industry etc. |
| 3. | To understand the concept of galvanic cellsand corrosion with theories like electro chemical theory. |
| 4. | To understand the importance of water. |
| 5. | To understand about the materials which are used in major industries like steel and metallurgical manufacturing industries, construction and electrical equipment manufacturing industries. |

**Course Outcomes:**At the end of the course students will be able to

| S.no | Out Come | KL |
|---|---|---|
| 1. | Develop polymer composites, synthetic polymers and formulation of polymers and their use in design | K3 |
| 2. | Apply the knowledge about quality of water and its treatment methods for domestic and industrial applications. Understanding the principle, mechanism of corrosion and utilization of various techniques to control. | K3 |
| 3. | Develop the knowledge of fuels and their economics, advantages and limitations. Make use of the basic concepts of semiconductors and liquid crystals for engineering applications. | K3 |
| 4. | Identify constituents of various ceramic materials, characteristics and their appropriate use in construction. Apply the knowledge of electrochemistry principles to design energy storage | K2 |

## SYLLABUS

| | |
|---|---|
| **UNIT-I (10Hrs)** | **High Polymers and Plastics; Rubbers & Elastomers**<br>Polymerization Definition, Types of Polymerization, free radical Mechanism of addition polymerization, Plastics as engineering materials, Thermoplastics and Thermosetting plastics, Compounding of plastics, Fabrication of plastics (4 techniques); Preparation, Properties and applications of Polyethylene, PVC, Bakelite, Nylon - 6,6,<br>Bullet Proof plastics -polycarbonate and Kelvar; Fiber reinforced plastics, conducting polymers,<br>Biodegradable Polymers - PHBV, Nylon 2, Nylon 6.<br>Natural rubber – Vulcanization – Compounding of Rubber; Preparation, properties and applications of Buna – S; Buna – N; |
| **UNIT-II (10Hrs)** | **Energy Sources and Applications:**<br>**Nuclear Energy:** Nuclear fission and Nuclear fusion – Nuclear Power reactor – Applications. Thermal fuels – Introduction – Classification – Calorific value – HCV and |

| | |
|---|---|
| | LCV – Bomb calorimeter; Coal : Proximate and ultimate analysis of coal – Significance of the analysis – Manufacture of coke by Otto Hoffman's by Product Process , Refining crude oil; Knocking; Chemical structure-Knocking, Octane number of gasoline, Cetane number of diesel oil, synthetic Petrol; LPG, CNG |
| | |
| **UNIT-III (12Hrs)** | **Electrochemical cells and Corrosion**<br>Galvanic cell, single electrode potential, Calomel electrode; Modern batteries: - Lead – Acid battery; Fuel cells- Hydrogen – Oxygen fuel cell, Lithium battery Theories of corrosion (i) dry Corrosion (ii) wet corrosion. Types of corrosion - differential aeration corrosion, pitting corrosion, galvanic corrosion, stress corrosion, Factors influencing corrosion, Protection from corrosion-material selection & design, cathodic protection, Protective coatings- metallic coatings – Galvanizing, Tinning, Electroplating; Electrrolessplating ;Paints |
| | |
| **UNIT-IV (8Hrs)** | **Water technology**<br>Sources of water – Hardness of water – Estimation of hardness of water by EDTA method; Boiler troubles – sludge and scale formation, Boiler corrosion, caustic embrittlement, Priming and foaming; Softening of water by Lime – Soda Process, Zeolite Process, Ion – Exchange Process; Municipal water treatment; Desalination of sea water by Electrodialysis and Reverse osmosis methods. |
| | |
| **UNIT-V (10Hrs)** | **Chemistry of Engineering Materials& Advanced Engineering materials**<br>**Cement:-** Manufacture of Portland cement, setting and hardening of cement, Deterioration of cement concrete.<br>**Refractories:-** Definition, Characteristics, classification, Properties and failure of refactories.<br>**Solar Energy: -** Construction and working of Photovoltaic cell, applications.<br>**Solid State Materials:** Crystal imperfections, Semi Conductors, Classification and chemistry of semi conductors: Intrinsic semiconductors; Extrinsic semiconductors; Defect semiconductors, Compound Semiconductors and Organic Semiconductors.<br>**Liquid Crystals: -** Definition – Classification with examples – Applications |

| | |
|---|---|
| **Text Books:** | |
| 1. | Engineering Chemistry by Jain and Jain, Dhanpat Rai Publishing co. |
| 2. | Engineering Chemistry by Willy India Pvt Ltd. |
| 3. | Engineering Chemistry by Dr.K.Anji Reddy and Dr.M.S.R.Reddy ; Silicon Publications. |
| | |
| **Reference Books:** | |
| 1. | Engineering Chemistry by Shikha Aharwal; Cambridge University Press, 2015 edition. |
| 2. | A text of Engineering Chemistry by S.S.Dara; S.Chand& Co Ltd. |
| 3. | Chemistry in Engineering and Technology by JC Kuriacose and J. Rajaram Mc. Graw Hill edition. |

| Course Code | Category | L | T | P | C | I.M | E.M | Exam |
|---|---|---|---|---|---|---|---|---|
| **B20IT1201** | **ES** | **3** | **--** | **--** | **3** | **30** | **70** | **3 Hrs.** |

# DIGITAL LOGIC DESIGN

(Common to AIDS & IT)

**Course Objectives:** The students are expected to learn about:

| | |
|---|---|
| 1. | To study the basic philosophy underlying the various number systems, negative number representation, binary arithmetic, theory of Boolean algebra and map method for minimization of switchingfunctions. |
| 2. | To introduce the basic tools for design of combinational and sequential digitallogic. |
| 3. | To learn simple digital circuits in preparation for computerengineering. |

**Course Outcomes:** At the end of the course students will be able to

| S.No | Outcome | KL |
|---|---|---|
| 1. | Demonstrate different number systems, binary addition andsubtraction, 2's complement representation and operations with thisrepresentation. | K2 |
| 2. | Understand the different switching algebra theorems and apply themfor logicfunctions. | K3 |
| 3. | Define the Karnaugh map for a few variables and make use for an algorithmic reduction of logicfunctions. | K3 |
| 4. | Understand various logic gates starting from simple ordinary gates to complex programmable logic devices &arrays and design different combinational logic circuits. | K3 |
| 5. | Design various sequential circuits starting from flip-flop to registers andcounters. | K3 |

## SYLLABUS

| | |
|---|---|
| **UNIT-I (10 Hrs)** | **Digital Systems and Binary Numbers**<br>Digital Systems, Binary Numbers, Octal and Hexadecimal Numbers, Complements of Numbers, Signed Binary Numbers, Arithmetic addition and subtraction, 4-bit codes: BCD, EXCESS 3, alphanumeric codes, 9's complement, 2421, etc. |
| **UNIT-II (10 Hrs)** | **Concept of Boolean algebra**<br>Basic Theorems and Properties of Boolean algebra, Boolean Functions, Canonical and Standard Forms, Minterms and Maxterms. Gate level Minimization Map Method, Three-Variable K-Map, Four Variable K-Maps. Products of Sum Simplification, Sum of Products Simplification, Don't – Care Conditions, NAND and NOR Implementation, Exclusive-OR Function. |
| **UNIT-III (12 Hrs)** | **Combinational Logic**<br>Introduction, Analysis Procedure, Binary Adder–Subtractor, Binary Multiplier, Decoders, Encoders, Multiplexers, Demultiplexers, Priority Encoder, Code Converters, Magnitude Comparator, HDL Models of Combinational Circuits.<br>Realization of Switching Functions Using PROM, PAL and PLA. |

| | |
|---|---|
| **UNIT-IV (10 Hrs)** | **Synchronous Sequential Logic**<br>Introduction to Sequential Circuits, Storage Elements: Latches, Flip-Flops, RS- Latch Using NAND and NOR Gates, Truth Tables. RS, JK, T and D Flip Flops, Truth and Excitation Tables, Conversion of Flip Flops |
| | |
| **UNIT-V (10 Hrs)** | **Registers and Counters**<br>Registers, Shift Registers, Ripple Counters, Synchronous Counters, Ring Counter, Johnson Counter. |
| | |

**Text Books:**

| | |
|---|---|
| 1. | Digital Design, 5/e, M.Morris Mano, Michael D Ciletti, PEA. |
| 2. | Fundamentals of Logic Design, 5/e, Roth, Cengage. |

**Reference Books:**

| | |
|---|---|
| 1. | Digital Logic and Computer Design, Morris Mano, Pearson India |
| 2. | Digital Logic Design, Brain Holds worth, Cline woods, O'Relly |
| 3. | Modern Digital Electronics, R.P Jain, Mc Graw Hill, 4th edition |

| Course Code | Category | L | T | P | C | I.M | E.M | Exam |
|---|---|---|---|---|---|---|---|---|
| **B20IT1202** | **ES** | **3** | **--** | **--** | **3** | **30** | **70** | **3 Hrs.** |

## OBJECT ORIENTED PROGRAMMING THROUGH C++

(Common to AIDS & IT)

**Course Objectives:**

| | |
|---|---|
| 1. | Describe the procedural and object oriented paradigm. |
| 2. | Understand dynamic memory management techniques using pointers, constructors, destructors, etc. |
| 3. | Describe the concepts of streams, classes, functions, data and object function overloading, operator overloading, virtual functions and polymorphism. |
| 4. | Classify inheritance with the understanding of early and late binding, usage of exception handling, generic programming. |
| 5. | Demonstrate the use of various OOPs concepts with the help of programs. |

**Course Outcomes**: At the end of the course, the student will be able to

| S.No | **Outcome** | **KL** |
|---|---|---|
| 1. | Differentiate between the procedural and object oriented paradigm. | K4 |
| 2. | Design object oriented applications using dynamic memory management techniques and overloading concepts. | K3 |
| 3. | Demonstrate inheritance, pointer, polymorphism and virtual functions | K2 |
| 4. | Apply implement generic programming, Exception handling in real time applications. | K3 |

## SYLLABUS

| | |
|---|---|
| **UNIT-I (10 Hrs)** | Introduction to C++: Difference between C and C++, Evolution of C++, The Object Oriented Technology, Disadvantage of Conventional Programming-, Key Concepts of Object Oriented Programming, Advantage of OOP, Object Oriented Language. |
| **UNIT-II (10 Hrs)** | Classes and Objects &Constructors and Destructor: Classes in C++-Declaring Objects, Access Specifiers and their Scope, Defining Member Function-Overloading Member Function, Nested class, Constructors and Destructors, Introduction, Constructors and Destructor- Characteristics of Constructor and Destructor, Application with Constructor, Constructor with Arguments (parameterized Constructor, Destructors- Anonymous Objects. |
| **UNIT-III (10 Hrs)** | Operator Overloading and Type Conversion & Inheritance: The Keyword Operator, Overloading Unary Operator, Operator Return Type, Overloading Assignment Operator (=), Rules for Overloading Operators,Inheritance, Reusability, Types of Inheritance, Virtual Base Classes, Object as a Class Member, Abstract Classes, Advantages of Inheritance-Disadvantages of Inheritance. |
| **UNIT-IV (8 Hrs)** | Pointers & Binding Polymorphisms and Virtual Functions: Pointer, Features of Pointers, Pointer Declaration, Pointer to Class, Pointer Object, The this Pointer, Pointer to Derived Classes and Base Class, Binding Polymorphisms and Virtual Functions, Binding in C++, Virtual Functions, Rules for Virtual Function, Virtual Destructor. |

| | |
|---|---|
| **UNIT-V (12 Hrs)** | Generic Programming with Templates, Need for Templates, Definition of class Templates, Normal Function Templates, Overloading of Template Function, Bubble Sort Using Function Templates, Difference Between Templates and Macros, Linked Lists with Templates, Exception Handling, Principles of Exception Handling, The Keywords try throw and catch, Multiple Catch Statements – Specifying Exceptions. |

**Text Books:**

| | |
|---|---|
| 1. | A First Book of C++, Gary Bronson, Cengage Learning. |
| 2. | The Complete Reference C++, Herbert Schildt, TMH. |

**Reference Books:**

| | |
|---|---|
| 1. | Object Oriented Programming C++, Joyce Farrell, Cengage. |
| 2. | C++ Programming: from problem analysis to program design, DS Malik, Cengage Learning. |
| 3. | Programming in C++, Ashok N Kamthane, Pearson 2nd Edition. |

**e-Resources**

| | |
|---|---|
| 1. | https://nptel.ac.in/courses/106/105/106105151/ |
| 2. | https://github.com/topics/object-oriented-programming |

| Course Code | Category | L | T | P | C | I.M | E.M | Exam |
|---|---|---|---|---|---|---|---|---|
| **B20EE1203** | **ES** | **3** | **--** | **--** | **3** | **30** | **70** | **3 Hrs.** |

## PRINCIPLES OF ELECTRICAL AND ELECTRONICS ENGINEERING

(Common to AIDS & IT)

**Course Objectives:**

| | |
|---|---|
| 1. | To learn the fundamentals of electrical circuits and single phase AC circuits |
| 2. | To understand the principle of operation and construction details of DC machines. |
| 3. | To understand the principle of operation of Transformers and AC machines. |
| 4. | To understand the basic operation of semiconductors devices. |
| 5. | To understand the basic operations of transistors BJT & FET |

**Course Outcomes:**At the end of the course, the student will be able to

| S.No | Out Come | KL |
|---|---|---|
| 1. | Apply concepts of Ohm’s Law, Kirchhoff's laws for solving DC circuits. | K3 |
| 2. | Apply Phasor representation concept to Analyze single-phase AC circuitsConsisting of series RL - RC - RLC combinations. | K3 |
| 3. | Apply the Faraday's laws and induced EMF concepts to describe the operating Principles and characteristics of DC Machines, Transformers and Induction motors. | K4 |
| 4. | Understand and apply basic concepts of semiconductors devices such as PN Junction diode and Zener diode. | K3 |
| 5. | Understand and Analyze the characteristics of BJT in CE, CB, CC configurationsand Analyze the characteristics of JFET. | K4 |

**SYLLABUS**

| | |
|---|---|
| **UNIT-I (12Hrs)** | **Introduction to Electrical circuits:** Introduction to electrical concepts. Types of network elements, Ohms Law, Kirchhoff’s Laws, Series and parallel Circuits connection of resistances with DC excitation. Representation of sinusoidal waveforms - Peak, Average and RMS values - Phasor representation - real power -Reactive power - apparent power - power factor. Analysis of single-phase ac circuits consisting of RL - RC - RLC series circuits |
| **UNIT-II (10Hrs)** | **D.C Machines:** Faraday's Laws-Induced EMF - Principle of operation of DC Generator - EMF equation – Construction-Types of DC generator-DC motor types, Torque equation –Losses. |
| **UNIT-III ( 8 Hrs )** | **Transformers and AC machines:** Principle of operation of single-phase transformer - EMF equation - equivalent circuit – losses– Transformer efficiency. principle and operation of Induction Motor [Elementary treatment only]. |
| **UNIT-IV (12 Hrs)** | **Semiconductors and P-N junction diode:** Types of semiconductors. Basic operation and V-I Characteristics of semiconductor diode, Operation of PN junction diode as rectifier, Avalanche breakdown and Zener breakdown phenomenon, Operation of Zener diode as regulator. Light Emitting Diode (LED). |

| | |
|---|---|
| **UNIT-V (8Hrs)** | **Bipolar Junction Transistor (BJT) and Field Effect Transistors (FET):** Introduction, construction, basic operation of NPN and PNP transistors, Transistor circuit Configurations- CE, CB, and CC- Input and output Characteristics in various configurations, Transistor as an amplifier, Junction field Effect Transistors (JFET), JFET characteristics [Elementary treatment of **FET**only]. |

| | |
|---|---|
| **Text Books:** | |
| 1. | Principles of Electrical Engineering, V.K mehta, Rohit Mehta, S. Chand Publications. Revised Edition 2012 |
| 2. | Electrical Technology by Surinder Pal Bali, Pearson Publications. January 1, 2013 |
| 3. | Electronic Device and Circuits by Sanjeev Guptha, Dhanpatrai & Co. Pvt. Ltd. |
| 4. | Electronic Device and Circuits by K.Satya Prasad, VGS |
| | |
| **Reference Books:** | |
| 1. | Fundamentals of Electrical Circuits (Sixth Edition) by Charles K Alexander & Matthew Sadiku, Mc Graw Hill Education. |
| 2. | Fundamentals of Electrical Engineering and Electronics, by Theraja B.L, S Chand Multicolor edition (1 December 2006) |
| 3. | Basic Electrical Engineering, by J.B Gupta, K Kataria and Sons; Reprint 2013 edition |
| 4. | Integrated Electronics- Millman & Halkias, TMH |
| 5. | Circuit Theory Analysis and Synthesis, Abhijit Chakrabarti, Dhanpat Rai & Co (Pvt) Ltd, Educational & Technical Publishers |

| Course Code | Category | L | T | P | C | I.M | E.M | Exam |
|---|---|---|---|---|---|---|---|---|
| B20BS1208 | BS | -- | -- | 3 | 1.5 | 15 | 35 | 3Hrs |

## APPLIED CHEMISTRY LAB

(Common to AIDS,CE,EEE & ME)

**Course Objectives:**

| | |
|---|---|
| 1. | To investigate and understand Physical behaviour in the laboratory using scientific reasoning and logic and interpret the result of simple experiments and demonstration of chemical Principle and also evaluate the impact of chemical discoveries on how we view the world. |
| 2. | Effectively communicate experimental results and solutions to application problems through oral and written reports. |
| 3. | Recognize the classical ideas and chemical phenomena and also define and analyse the concepts. |

**Course Outcomes:** At the end of the course students will be able to

| S.No | Out Come | KL |
|---|---|---|
| 1. | Gain technical knowledge of measuring, operating and testing of chemical instruments and equipments. Carrying out different types of chemical reactions for analysing different materials in micro level quantities. | K3 |
| 2. | Analyze and generate experimental skills to enhance the analytical thinking capabilities in the modern trends in engineering and technology. | K3 |

## LIST OF PROGRAMS

| | |
|---|---|
| 1 | Determination of Alkalinity of water sample. |
| 2 | Determination of total hardness of water by EDTA method. |
| 3 | Estimation of Ferrous Iron by $KMnO_4$. |
| 4 | Estimation of oxalic acid by $KMnO_4$. |
| 5 | Estimation of Mohr's salt by $K_2Cr_2O_7$. |
| 6 | Estimation of Dissolved oxygen by Winkler's method. |
| 7 | Determination of pH of water and soil sample. |
| 8 | Determination of Chlorides present in water sample. |
| 9 | Conductometric titration of strong acid Vs strong base. |
| 10 | Potentiometric titration of strong acid Vs strong base. |
| 11 | Potentiometric titration of strong acid Vs weak base. |
| 12 | Preparation of Phenol formaldehyde resin. |
| 13 | Determination of saponification value of oils. |
| 14 | Determination of pour and cloud points of lubricating oil. |
| 15 | Determination of Acid value of oil. |

**Reference Books:**


| | |
|---|---|
| 1. | Engineering Chemistry Lab Manual Prepared by Chemistry Faculty of S.R.K.R.EngineeringCollege. |
| 2. | Laboratory manual on Engineering Chemistry by Dr.Sudha Rani; Dhanpat Rai Publishing Company. |
| 3. | Engineering Chemistry Laboratory manual – I & II by Dr.K.Anji Reddy; Tulip Publications. |

| Course Code | Category | L | T | P | C | I.M | E.M | Exam |
|---|---|---|---|---|---|---|---|---|
| B20HS1202 | HS | -- | -- | 3 | 1.5 | 15 | 35 | 3Hrs |

## COMMUNICATION SKILLS LAB

(Common to AIDS ,CE,CSE,ECE,EEE,IT & ME)

**Course Objectives:**

| | |
|---|---|
| 1. | To expose to a variety of self-instructional, learner-friendly modes of languagelearning. |
| 2. | To familiarize the students with CALL (Computer Assisted Language Learning). Thus, providing them with the required facility to face computer-based competitive exams like GRE,TOEFL, GMAT etc. |
| 3. | To equip the students with necessary professional communication. |
| 4. | To build confidence in LSRW Skills. |
| 5. | To adapt the students by adopting the techniques of effective communication skills. |

**Course Outcomes:**At the end of the course students will be able to

| S.No | Out Come | KL |
|---|---|---|
| 1. | Apply their linguistic competence in all LSRW skills to professional and personal settings. | K3 |
| 2. | Apply communication skills learnt through various language learning activities to their advancement in academics and competitive examinations. | K3 |
| 3. | Draft job application letters, E-Mail messages and other writing discourses. | K3 |
| 4. | Adopt professional etiquette consistent with formal settings. | K3 |
| 5. | Improve fluency and clarity in both spoken and written English. | K3 |

## SYLLABUS

| | |
|---|---|
| **UNIT-I** | A list of communicative expressions<br>(Requests, Permissions, Asking/ giving directions, Thanking and Responding to Thanks, Clarifying, Inviting, Congratulating, Advising, Agreeing and disagreeing etc.,)<br>Common Errors |
| **UNIT-II** | Pronunciation Letters and Sounds<br>The Sounds of English<br>Stress and Intonation<br>Phonetic Transcription |
| **UNIT-III** | Group Discussions |
| **UNIT-IV** | Presentation Skills |
| **UNIT-V** | Interview Skills<br>Resume/ Curriculum Vitae<br>Covering Letter<br>FAQ's<br>Telephonic Interviews/ Etiquette<br>Mock Interviews |

| **Text Books:** | |
|---|---|
| 1. | Interact – English Lab Manual for Undergraduate Students – Orient BlackSwan |
| | |
| **Reference Books:** | |
| 1. | Exercises in Spoken English Part 1,2,3,4, OUP and CIEFI. |
| 2. | English Pronunciation in use- Mark Hancock, CUP. |
| 3. | English Pronunciation in use- Mark Hewings, CUP. |
| 4. | English Pronunciation Dictionary- Daniel Jones, CUP. |
| 5. | English Phonetics for Indian Students- P. BalaSubramanian, Mac MillanPublications |
| 6. | Technical Communication- Meenakshi Raman, Sangeeta Sharma, OUP. |
| 7. | Technical Communication- Gajendra Singh Chauhan, SmitaKashiramka, cengage Publications |

| Course Code | Category | L | T | P | C | I.M | E.M | Exam |
|---|---|---|---|---|---|---|---|---|
| B20IT1203 | ES | -- | -- | 3 | 1.5 | 15 | 35 | 3 Hrs. |

## OBJECT ORIENTED PROGRAMMING THROUGH C++ LAB

(Common to AIDS & IT)

**Course Objectives:**

| | |
|---|---|
| 1. | To Exploit object oriented procedure of solving problems the basic principles and benefits of object-oriented procedure of solving problems in the context of C++ language. |
| 2. | To aware and write programs for generic programming concepts. |

**Course Outcomes:** At the end of the course students will be able to

| S.No | Outcome | KL |
|---|---|---|
| 1 | Apply the basic concepts in C++ like Class and objects | K3 |
| 2 | Analyze memory management techniques like constructor, destructor and overloading mechanisms | K4 |
| 3 | Apply reusability of code and usage of exception handling and generic programming | K3 |

## LIST OF EXPERIMENTS

| | |
|---|---|
| 1. | Write a C++ Program to display Names, Roll No., and grades of 3 students who have appeared in the examination. Declare the class of name, Roll No. and grade. Create an array of class objects.Read and display the contents of the array. |
| 2. | Write a C++ program to declare Struct. Initialize and display contents of member variables. |
| 3. | Write a C++ to illustrate the static methods and members. |
| 4. | Write a C++ program to find the sum for the given variables using function with default arguments. |
| 5. | Write a C++ Program to illustrate Enumeration and Function Overloading |
| 6. | Write a C++ program to declare a class. Declare pointer to class. Initialize and display the contents of the class member. |
| 7. | To write a C++ program to find the value of a number raised to its power that demonstrates a function using call by value. |
| 8. | To write a C++ program and to implement the concept of Call by Address |
| 9. | Given that an EMPLOYEE class contains following members: data members: Employee number, Employee name, Basic, DA, IT, Net Salary and print data members. |
| 10. | Write a C++ program to read the data of N employee and compute Net salary of each employee (DA=52% of Basic and Income Tax (IT) =30% of the gross salary). |
| 11. | Write a C++ to illustrate the concepts of console I/O operations. |
| 12. | Write a C++ program to use scope resolution operator. Display the various values of the same variables declared at different scope levels. |
| 13. | Write a C++ program to allocate memory using new operator. |
| 14. | Write a C++ program to create multilevel inheritance. (Hint: Classes A1, A2, A3) |
| 15. | Write a C++ program to create an array of pointers. Invoke functions using array objects. |
| 16. | Write a C++ program to use pointer for both base and derived classes and call the member function. Use Virtual keyword |
| 17. | Write a C++ program for swapping two values using function templates |

| **Reference Books:** | |
|---|---|
| 1. | A First Book of C++, Gary Bronson, Cengage Learning. |
| 2. | The Complete Reference C++, Herbert Schildt, TMH. |

| Course Code | Category | L | T | P | C | I.M | E.M | Exam |
|---|---|---|---|---|---|---|---|---|
| **B20MC1201** | **MC** | **2** | **--** | **--** | **0** | **--** | **--** | **--** |

## ENVIRONMENTAL SCIENCE

(Common to AIDS,CE,CSBS,EEE & ME)

**Course Objectives:**The objectives of the course are to impart:

| | |
|---|---|
| 1. | Overall understanding of the natural resources. |
| 2. | Basic understanding of the ecosystem and its diversity. |
| 3. | Acquaintance on various environmental challenges induced due to unplanned anthropogenic activities. |
| 4. | An understanding of the environmental impact of developmental activities. |
| 5. | Awareness on the social issues, environmental legislation and global treaties. |

| **Course outcomes :** After completion of the course, students will be able to | | **K L** |
|---|---|---|
| 1 | Bring awareness among the students about the nature and natural ecosystems | K2 |
| 2 | Sustainable utilization of natural resources like water, land, energy and air | K4 |
| 3 | Resource pollution and over exploitation of land, water, air and catastrophic (events) impacts of climate change, global warming, ozone layer depletion, marine, radioactive pollution etc to inculcate the students about environmental awareness and safe transfer of our mother earth and its natural resources to the next generation | K5 |
| 4 | Constitutional provisions for the protection of natural resources | K2 |
| 5 | Green technologies and its applications | K3 |

## SYLLABUS

| | |
|---|---|
| **UNIT-I (8 Hrs)** | **Multidisciplinary nature of Environmental Studies:** Definition, Scope and Importance – Sustainability: Stockholm and Rio Summit–Global Environmental Challenges: Global warming and climate change, acid rains, ozone layer depletion, population growth and explosion, effects;. Role of information technology in environment and human health.<br>**Ecosystems:** Concept of an ecosystem. - Structure and function of an ecosystem; Producers, consumers and decomposers. - Energy flow in the ecosystem - Ecological succession. - Food chains, food webs and ecological pyramids; Introduction, types, characteristic features, structure and function of Forest ecosystem, Grassland ecosystem, Desert ecosystem, Aquatic ecosystems. |
| | |
| **UNIT-II (8 Hrs)** | **Natural Resources: Natural resources and associated problems.**<br>Forest resources: Use and over – exploitation, deforestation – Timber extraction – Mining, dams and other effects on forest and tribal people.<br>Water resources: Use and over utilization of surface and ground water – Floods, drought, conflicts over water, dams – benefits and problems.<br>Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources.<br>Food resources**:** World food problems, changes caused by non-agriculture activities-effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity.<br>Energy resources: Growing energy needs, renewable and non-renewable energy sources use of alternate energy sources. |

| | |
|---|---|
| | Land resources: Land as a resource, land degradation, Wasteland reclamation, man induced landslides, soil erosion and desertification; Role of an individual in conservation of natural resources; Equitable use of resources for sustainable lifestyles. |
| | |
| **UNIT-III (8 Hrs)** | **Biodiversity and its conservation:** Definition: genetic, species and ecosystem diversity-classification - Value of biodiversity: consumptive use, productive use, social-Biodiversity at national and local levels. India as a mega-diversity nation - Hot-sports of biodiversity - Threats to biodiversity: habitat loss, man-wildlife conflicts. - Endangered and endemic species of India – Conservation of biodiversity: conservation ofbiodiversity. |
| | |
| **UNIT-IV (8 Hrs)** | **Environmental Pollution:** Definition, Cause, effects and control measures of Air pollution, Water pollution, Soil pollution, Noise pollution, Nuclear hazards. Role of an individual in prevention of pollution. - Pollution case studies, Sustainable Life Studies. Impact of Fire Crackers on Men and his wellbeing.<br>**Solid Waste Management:** Sources, Classification, effects and control measures of urban and industrial solid wastes. Consumerism and waste products, Biomedical, Hazardous and e – waste management. |
| | |
| **UNIT-V (8 Hrs)** | **Social Issues and the Environment:** Urban problems related to energy -Water conservation, rain water harvesting-Resettlement and rehabilitation of people; its problems and concerns. Sustainability: theory and practice, Environmental ethics: Issues and possible solutions. Environmental Protection Act -Air (Prevention and Control of Pollution) Act. – Water (Prevention and control of Pollution) Act -Wildlife Protection Act -Forest Conservation Act-Issues involved in enforcement of environmental legislation.-Public awareness. |
| | |
| **UNIT-VI (8 Hrs)** | **Environmental Management**: Impact Assessment and its significance various stages of EIA, preparation of EMP and EIS, Environmental audit. Ecotourism, Green Campus – Green business and Greenpolitics. Environmental dairy.<br>**The student should Visit an Industry / Ecosystem and submit a report individually on any issues related to Environmental Studies course and make a power point presentation.** |

**Text Books:**

| | |
|---|---|
| 1. | Environmental Studies, K. V. S. G. Murali Krishna, VGS Publishers,Vijayawada Rani; Pearson Education, Chennai |
| 2. | Environmental Studies, R. Rajagopalan, 2$^{nd}$ Edition, 2011, Oxford UniversityPress. |
| 3. | Environmental Studies, P. N. Palanisamy, P. Manikandan, A. Geetha, andK.Manjula |

**Reference Books:**

| | |
|---|---|
| 1. | Text Book of Environmental Studies, Deeshita Dave & P. Udaya Bhaskar, CengageLearning. |
| 2. | A Textbook of Environmental Studies, Shaashi Chawla, TMH, NewDelhi |
| 3. | Environmental Studies, Benny Joseph, Tata McGraw Hill Co, NewDelhi |
| 4. | Perspectives in Environment Studies, Anubha Kaushik, C P Kaushik, New AgeInternational Publishers,2014 |

| Code | Category | L | T | P | C | I.M | E.M | Exam |
|---|---|---|---|---|---|---|---|---|
| **B20MC1203** | **MC** | **--** | **--** | **2** | **--** | **--** | **--** | **--** |

## NATIONAL SERVICE SCHEME(NSS)

(Common to All Branches)

**Course Objectives:**

| | |
|---|---|
| 1. | To understand the community and understand themselves in relation to their community. |
| 2. | Identify the needs and problems of the community and involve them in problem solving process. |
| 3. | Utilize their knowledge for finding practical solution to individual and community problems. |

**Course Outcomes: Student will be able to**

| S.No | | Knowledge Level |
|---|---|---|
| 1. | **understand** general orientation about community service, voluntarism role and responsibility of NSS volunteer. | K2 |
| 2. | **Analyze** about the community he live in. | K4 |
| 3. | **Asses** the life in adopted villages. | K5 |
| 4. | **Identify** the importance of national days and attain participation in it. | K3 |

## SYLLABUS

| | |
|---|---|
| 1. | Volunteerism- community and beyond(Theory). |
| 2. | Role and responsibility of NSS volunteer (Theory). |
| 3. | General orientation about community service(Theory). |
| 4. | Arranging lectures on social issues in schools or villages(Theory). |
| 5. | Arranging rally's on social issues. |
| 6. | Socio economic survey in adopted villages |
| 7. | Plantation of saplings. |
| 8. | Blood donation camp |
| 9. | Rainwater harvesting awareness camp. |
| 10. | Celebration of national days as per NSS list. |